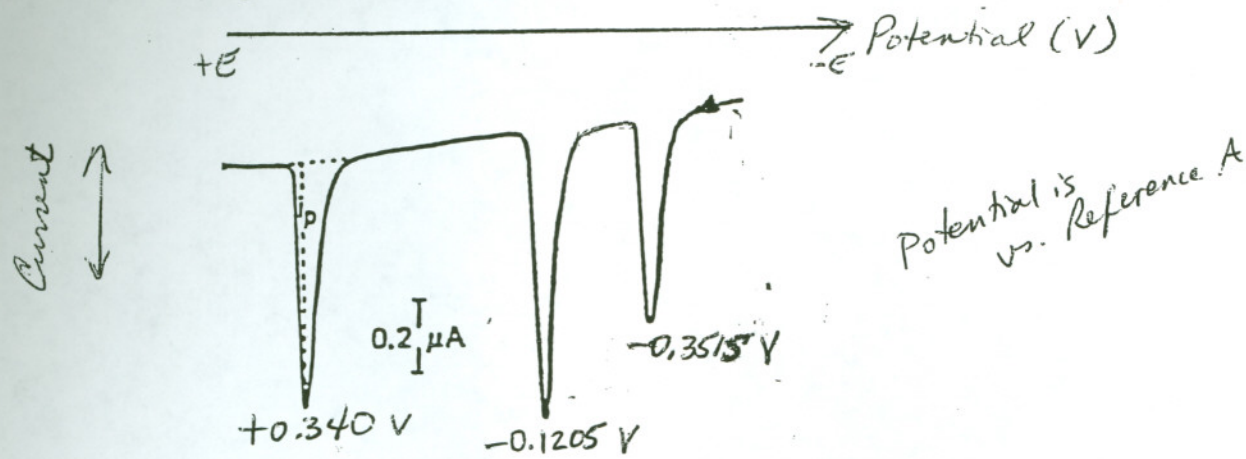
Analytical Chemistry Cumulative Exam
March 26, 2004
(Electroanalytical)

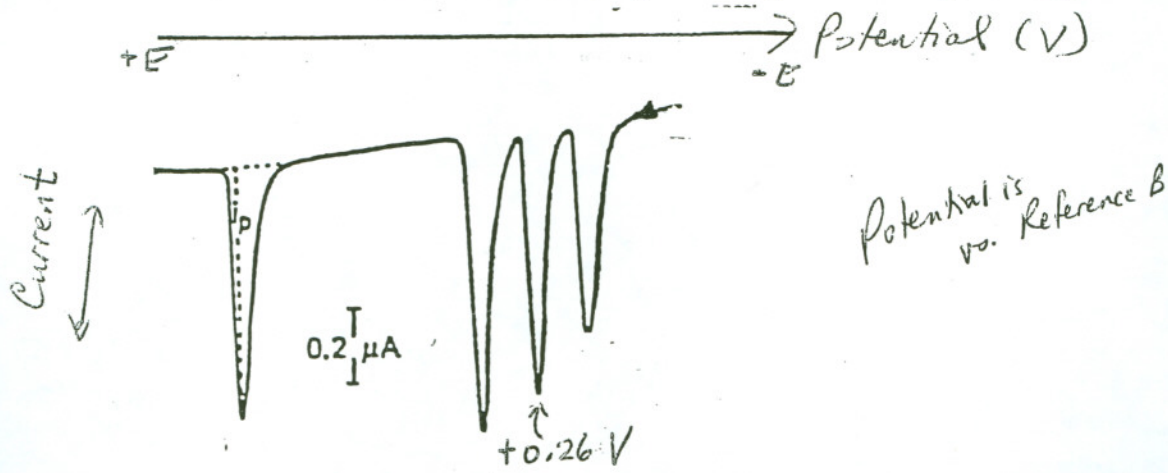
1. (**33 points**) A mixture of metals is analyzed by stripping voltammetry. A voltammogram is shown below.



a. Given the voltammogram above, identify the metals in the mixture using the following table of species with E° values (in volts) relative to Reference A.

Ag+ + e = Ag°	+0.7991
Cu^{2+} + 2e = Cu°	+0.340
Pb^{2+} + 2e = Pb°	-0.1205
In^{2+} + 2e = In°	-0.2344
Cd^{2+} + 2e = Cd°	-0.3515
Fe2+ + 2e = Fe°	-0.44
Zn^{2+} + 2e = Zn°	-0.7626

b. Results on another sample from a different lab reveals a similar-looking pattern of peaks, except now there is a new peak (see below). Unfortunately, this voltammogram was obtained using a different reference electrode, Reference B. You learn that Reference B is –0.500 V vs. Reference A. Identify the new peak. Show your work.



2. **(33 points)** A new organometallic molecule is synthesized. It is electroactive. You determine how many electrons are transferred per molecule by using a thin electrochemical cell, which has a known volume of 0.188 cm^3 (100 μm gap x 2.5 cm wide x 7.5 cm long). Integration of the current over the peak under the CV response (very slow time scales) is 3.63×10^{-2} Coulombs for a concentration of 1×10^{-3} M. What is n (the number of moles of electrons per mole of molecules)? Note that the Faraday constant is 96485 Coulombs/mol of electrons.

3. **(34 points)** A spoon with an area of 4 cm^2 is plated with a layer of an average thickness of 95.7 μm of silver from a Ag^+ solution (atomic weight of Ag = 107.8682 g/mol, density of Ag = 10.5 g/cm^3). What current needs to be applied to achieve this thickness in 2 hours if the current efficiency is 80%? Note that the Faraday constant is 96485 Coulombs/mol of electrons.